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* More implications of sub-model:

→ Partial correlation coefficient: $(Y = \beta_0 + \beta_1 X^{(1)} + \dots + \beta_{p-1} X^{(p-1)} + \varepsilon)$

↳ = Correlation b/w the residuals $e^{(-j)}$ & $r^{(-j),j}$.

(This is partial corrⁿ b/w Y & $X^{(j)}$)

Here, $e^{(-j)}$ is the ratio of residuals from fitting linear regression of Y in $\{X^{(k)}\}_{k \neq j}$

& $r^{(-j),j}$ is the ratio of residuals from fitting linear regression of $X^{(j)}$ on $\{X^{(k)}\}_{k \neq j}$.

$$e^{(-j)} = (I - H_{(-j)})Y \quad ; \quad r^{(-j),j} = (I - H_{(-j)})X^{(j)}$$

↑ orthogonal projection onto $\mathcal{C}([1: X^{(1)} \dots X^{(j-1)} X^{(j+1)} \dots X^{(p)}])$

$$\therefore \text{Corr}_{Y X^{(j)} | \{X^{(k)}\}_{k \neq j}} = \frac{(X^{(j)})^T (I - H_{(-j)}) Y}{\sqrt{Y^T (I - H_{(-j)}) Y} \cdot \sqrt{(X^{(j)})^T (I - H_{(-j)}) (X^{(j)})}}$$

• Inference question:

$$Y = X\beta + \varepsilon = Z\gamma + W\delta + \varepsilon \quad (\text{full model})$$

$$Y = Z\gamma + \varepsilon \quad (\text{submodel}) \quad [\delta = 0]$$

Q) Is the submodel adequate in describing the variability in Y ?

$$\rightarrow H_0: \delta = 0 \quad \text{v/s} \quad H_A: \delta \neq 0$$

$$\Leftrightarrow H_0: A\beta = 0 \quad \text{v/s} \quad H_A: A\beta \neq 0 \quad (A = [0: I])$$

- Anova decomposition:

$$Y^T Y = Y^T P_X Y + \underbrace{Y^T (I - P_X) Y}_{SSE}$$

$$= \underbrace{Y^T P_Z Y}_{SSR_{\text{sub}}} + \underbrace{Y^T P_{W|Z} Y}_{\text{additional variability explained by predictors } W} + Y^T (I - P_X) Y$$

To test $H_0: \delta = 0$ we may use the stat. $\frac{Y^T P_{W|Z} Y}{MSE}$

$$= \frac{Y^T P_{W|Z} Y}{Y^T (I - P_X) Y / (n - p(X))}$$

Alternatively, we may also use the test stat.

$$\hat{\delta}^T (\widehat{\text{Var}}(\hat{\delta}))^{-1} \hat{\delta}$$

$$\cdot \hat{\delta} = (W^T (I - P_Z) W)^{-1} W^T (I - P_Z) Y$$

$$\cdot \widehat{\text{Var}}(\hat{\delta}) = \sigma^2 (W^T (I - P_Z) W)^{-1}$$

$\cdot \hat{\delta} = A \hat{\beta} = (0: I) \hat{\beta}$ is the component of $\hat{\beta}$ corresponding to δ .

$$\rightarrow = \frac{1}{MSE} \hat{\delta}^T (W^T (I - P_Z) W) \hat{\delta}$$

$$= \frac{1}{MSE} Y^T (I - P_Z) W (W^T (I - P_Z) W)^{-1} (W^T (I - P_Z) W) (W^T (I - P_Z) W)^{-1} W^T (I - P_Z) Y$$

$$= \frac{1}{MSE} Y^T (I - P_Z) W (W^T (I - P_Z) W)^{-1} W^T (I - P_Z) Y$$

* Characterization of probability distributions:

Let X & Y be random vectors

$X \stackrel{D}{=} Y$ iff $\phi_X(t) = \phi_Y(t) \forall t$ where ϕ_X, ϕ_Y are characteristic funcⁿs of X & Y . $\phi_X(t) = E(e^{it^T X})$.

Let Ψ_X be the moment generating function of X .

~~if~~ $\phi_X(t) = \Psi_X(t) \forall t$

- If $\psi_X(t) < \infty$ in an open neighbourhood of 0, then $\phi_X(t) = \psi_X(it) \forall t$ in that nbd. and $\exists$ an analytic extension and the mgf uniquely identifies the distⁿ of X .
- M.G.F. of χ_n^2 : $\chi_n^2 = Z_1^2 + \dots + Z_n^2$, $Z_1, \dots, Z_n \stackrel{iid}{\sim} N(0,1)$
 $\chi_2^2 \sim \text{Exp}(\frac{1}{2})$, $\text{Exp}(\lambda) \sim \lambda e^{-\lambda x} (x > 0)$

Calculate MGF of χ_2^2 : $\psi_2(t) = \frac{1}{1-2t}$, $t < \frac{1}{2}$

$$\psi_2(t) = (\psi_1(t))^2$$

$$\Rightarrow \psi_1(t) = \frac{1}{\sqrt{1-2t}} \text{ for } t < \frac{1}{2}$$

$$\therefore \psi_n(t) = (\psi_1(t))^n$$

$$\therefore \text{MGF of } \chi_n^2 : \psi_n(t) = \begin{cases} (1-2t)^{-n/2} & \text{if } t < \frac{1}{2} \\ \infty & \text{o.w.} \end{cases}$$

- MGF of a non-central χ^2 random variable.

Sp. $Y \sim N_n(\mu, I_n)$, $\mu \in \mathbb{R}^n$

MGF of $Y^T A Y$ where A is a symm. n.d. matrix is: $|\det(I - 2tA)|^{-1/2} \exp(-\frac{1}{2}\mu^T \mu + \frac{1}{2}\mu^T (I - 2tA)^{-1} \mu)$ for t s.t. $(I - 2tA)^{-1}$ exists. (#check how)

Theorem: If $Y \sim N(\mu, I)$ then, for symm. A , $Y^T A Y \sim \chi_{P(A)}^2(\frac{1}{2}\mu^T A \mu)$ iff A is idempotent.

Proof: When $\mu = 0$ ($\Rightarrow$ direction)

mgf of $Y^T A Y$ is $\psi(t) = |\det(I - 2tA)|^{-1/2}$

If $r = P(A)$, mgf of χ_r^2 is $(1-2t)^{-r/2}$

By spectral decomposition of A , its ev's are $\lambda_1, \dots, \lambda_r, 0, \dots, 0$.

$$\begin{aligned}\psi(t) &= |\det(I - 2tA)|^{-1/2} \\ &= \left(\prod_{j=1}^r (1 - 2t\lambda_j) \right)^{-1/2}\end{aligned}$$

$\therefore Y^T A Y \sim \chi^2_{P(A)}$, we have.

$$\Rightarrow (1 - 2t)^{-r/2} = \left(\prod_{j=1}^r (1 - 2t\lambda_j) \right)^{-1/2}$$

$$\Rightarrow (1 - 2t)^r = \prod_{j=1}^r (1 - 2t\lambda_j)$$

$$\Rightarrow \lambda_j = 1 \quad \forall j = 1, \dots, r$$

$$\Rightarrow A \text{ is idempotent}$$

* Fisher-Cochran Theorem:

Sp. $Y \sim N_n(\mu, I)$ and $A_1, \dots, A_r$ are symm. $n \times n$, positive semidefinite matrices s.t.

$$Y^T Y = \sum_{j=1}^r Y^T A_j Y \quad \text{-- (i)}$$

Then, $Y^T A_j Y \sim \chi^2_{P(A_j)}(\frac{\lambda_j}{2})$ for $j = 1, \dots, r$ and

~~if~~ these ~~qud~~ quadratic forms are independent iff $\sum_{j=1}^r P(A_j) = n$ and in that case

$$\begin{aligned}\lambda_j &= \mu^T A_j \mu \quad \text{for } j = 1, \dots, r \\ &\& \sum_{j=1}^r \lambda_j = \mu^T \mu.\end{aligned}$$

Furthermore we then have $A_j A_k = 0$ for $j \neq k$ and A_j 's are idempotent.

Proof: $Y^T A_j Y \sim \chi^2_{P(A_j)} \left(\frac{\lambda_j}{2} \right) \Rightarrow \exists X_j, j=1, \dots, r$
 s.t. $X_j \sim N_{P(A_j)}(\mu_j, I)$ w/ $Y^T A_j Y \stackrel{D}{=} X_j^T X_j$ & $\lambda_j = \mu_j^T \mu_j$

Outline of Proof:

$$A_j = \underbrace{C_j C_j^T}_{n \times P(A_j)}$$

$$\therefore Y^T Y = \sum_{j=1}^r (C_j^T Y)^T (C_j^T Y)$$

$$= Y^T \left(\sum_{j=1}^r C_j C_j^T \right) Y$$

$$= Y^T B Y \quad \forall Y \in \mathbb{R}^n$$

$\Rightarrow$ All ev's of B are 1

$$B = \underbrace{[C_1 \dots C_r]}_{n \times \sum P(A_j)} \begin{bmatrix} C_1^T \\ \vdots \\ C_r^T \end{bmatrix}$$

$\therefore \sum P(A_j) = n$, C is $n \times n$

$$\Rightarrow I_n = C^T C = \begin{bmatrix} C_1^T \\ \vdots \\ C_r^T \end{bmatrix} [C_1 \dots C_r] = ((C_i^T C_j))_{\substack{1 \leq i \leq n \\ 1 \leq j \leq n}}$$

$$\parallel \begin{bmatrix} I_{P(A_1)} & 0 \\ 0 & I_{P(A_r)} \end{bmatrix}$$

~~APP~~ APPARENTLY